Dimitri Chrysafis

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Education

University of Wisconsin–Madison August 2025 - May 2027

Bachelor of Science in Computer Science

Relevant Coursework: COMP SCI 577 (Algorithms), COMP SCI 524 (Optimization), COMP SCI 400 (Programming III), COMP SCI 300 (Programming II), COMP SCI 252 (Computer Engineering), MATH 435 (Cryptography), MATH 331 (Probability), MATH 340 (Matrix and Linear Algebra)

Technical Skills

Languages: Python, C++, Julia, Go, JavaScript

ML/GPU: PyTorch, CatBoost, WebGPU/WGSL, WebGL/GLSL, Apple Metal, inference, quantization, model serving

Experience

Quantum Circuit Routing [Final Report] **COMPSCI 524 Project** Jul 2026

- Built an exact Julia/JuMP/HiGHS MILP for routing quantum circuits on sparse superconducting hardware
- Solved 24 instances to proven optimality; grid3x3 lowered the objective **56.1%** versus heavy-7

Projects

Qwen3.8-Flash-Next NVMe Inference Engine [GitHub]

- Building a disk-backed inference engine that runs the **180B-parameter Qwen3.8-Flash-Next** model on a Mac with **36 GB unified memory** by streaming MoE experts from NVMe instead of loading the full model into memory
- Implementing memory-mapped weights, dynamic expert caching, and asynchronous prefetching; benchmarking **throughput, latency, disk I/O, and peak memory**

Interactive 3D Fluid Simulator (MLS-MPM) [Demo] [GitHub]

- Built a browser-based 3D MLS-MPM simulator in JavaScript/WebGPU, keeping particle and grid state in GPU storage buffers and simulating **400,000 particles** with staged compute kernels
- Implemented fixed-point atomic accumulation for particle-to-grid transfers plus interactive dam-break, piston, particle-injection, and orbit-camera controls

Tennis Match Prediction System

- Built a feature pipeline over **92,000 ATP matches (1982–2024)** with player and surface-specific ELO, head-to-head, recent-form, and schedule-fatigue features
- Trained a CatBoost ensemble to predict winners; achieved **83.7% accuracy** and outperformed a baseline model

GPU Fractal Rendering Engine [GitHub] [Blog]

- Built GPU fractal renderers in WebGL2/GLSL and Apple Metal for Newton, Kleinian, and Julia sets with interactive pan/zoom and shader-based iteration
- Implemented a Metal compute pipeline for **6,000×6,000** Julia frames with adaptive iteration counts and GPU-parallel image generation

GPU-Accelerated Sphere Packing in Arbitrary Meshes [GitHub] [Blog]

- Built a stochastic tangent-sphere packing solver for arbitrary OBJ meshes, enforcing interior, surface-clearance, and pairwise non-overlap constraints
- Accelerated ray-mesh containment, nearest-surface queries, and candidate search with batched PyTorch tensors on MPS/CUDA; visualized results in Open3D